MINYOUNG KIM

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RESEARCH INTERESTS

My research lies at the intersection of 3D graphics, XR, human-agent interaction, and multimodal AI. I develop computational methods for synthesizing interactive XR experiences, virtual agents, and spatially grounded behaviors using scene understanding and large multimodal models.

EDUCATION

George Mason University 2021 - *Current*
Ph.D. Student in Computer Science (Advisor: Dr. Lap-Fai (Craig) Yu)

Ewha Womans University 2019 - 2021
M.S. in Computer Science and Engineering (Advisor: Dr. Young J. Kim)

Keimyung University 2012 - 2017
B.E. in Game and Mobile Engineering

PUBLICATIONS

Role-Aware Virtual Agents for Navigational Interaction Guided by a Multimodal Large Language Model
Minyoung Kim, Changyang Li, Cuong Nguyen, Lap-Fai Yu
ACM Transactions on Graphics (Proceeding of SIGGRAPH) 2026, Accepted [Project]

Crafting Dynamic Virtual Activities with Advanced Multimodal Models,
Changyang Li, Qingan Yan, **Minyoung Kim**, Zhan Li, Yi Xu, Lap-Fai Yu,
IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2025 [Project] [Paper]

Dragon's Path: Synthesizing User-Centered Flying Creature Animation Paths for Outdoor Augmented Reality Experiences
Minyoung Kim, Rawan Alghofaili, Changyang Li, Lap-Fai Yu
SIGGRAPH 2024 Conference Papers [Project] [Paper]

Location-Aware Adaptation of Augmented Reality Narratives
Wanwan Li*, Changyang Li*, Minyoung Kim, Haikun Huang, Lap-Fai Yu
Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems [Project] [Paper]

Synthesizing Human Faces using Latent Space Factorization and Local Weights
Minyoung Kim, Young J. Kim
In 2021 Computer Graphics International Conference (CGI) [Project] [Paper]

3D Surface Painting in VR using Force Feedback
Minyoung Kim, Young J. Kim
Journal of the Korea Computer Graphics Society, 2020 [Project] [Paper]

RESEARCH EXPERIENCE

Dolby Laboratories Sunnyvale, California
Research Intern (Advisor: Mark Hollar) May 2026 - Aug 2026
Project - "Performance to Scene"

- Developing AI-driven methods for translating human performance into spatially grounded cinematic representations.

- Building multimodal pipelines for human motion understanding in XR/media applications.

George Mason University Fairfax, Virginia
Graduate Research Assistant (Advisor: Dr. Lap-Fai (Craig)) Aug 2021 - Present

- Led research on XR experience synthesis, virtual agents, and spatially adaptive AR narratives.
- Developed systems for user-centered creature animation paths and multimodal role-aware virtual agents.
- Published first-author work at SIGGRAPH 2024 and ACM SIGGRAPH 2026.

Ewha Womans University Seoul, South Korea
Research Intern (Advisor: Dr. Young J. Kim) Jan 2019 - Feb 2019
Project - "Haptic Painting in VR" and "Human Face Geometry Processing with GNN"

- Prototyped a haptic VR system providing force feedback and collision detection between virtual objects and users.
- Developed a GNN-based human face generative model, resulting in a first-author CGI publication.

SKILLS

Programming Python, C++, C#, Java, JavaScript

ML/AI PyTorch, multimodal LLMs, computer vision, deep learning

Graphics/XR Unity, Unreal Engine, OpenGL, 3D animation, AR/VR prototyping

Tools/Devices Git, LaTeX, Blender, Hololens 2, Meta Quest

WORK EXPERIENCE

Nova Mobile System Carlsbad, CA, USA
Junior Mobile Programmer Intern - "Nova Mobile App" May 2017 - Apr 2018

Qualcomm Institute San Diego, CA, USA
Student Intern - "Health-Related IoT Tracking Project" Jan 2016 - Feb 2016

SERVICES

Conference Paper Reviewer SIGGRAPH 2025

Student Volunteer SIGGRAPH ASIA 2022 in Daegu

TALKS

Invited Seminar, Hong Kong Metropolitan University (HKMU) Oct, 2024

Invited Talk, GAMES (Graphics And Mixed Environment Symposium) Jul, 2024

Demo, ABC News [Video] May, 2023

AWARDS

Best Paper Award, Korea Computer Graphics Society (KCGS) 2019